

**SGGS Institute of Engineering and Technology, Vishnupuri, Nanded****Summer-Term Examination 2024-25****T.Y. B. Tech.****Chemical Engineering****Process Dynamics and Control****PCC-CH321****Date: 17.06.2025****Time: 10.00 am - 01.00 pm****Max. Marks: 100**

- Note:
- Attempt all questions
 - Assume suitable data if necessary
 - Use of non-programmable calculator is permitted.

Q.N.	Question	Marks
Q.1	What do you mean by time constant for first order system and derive transfer function for temperature measurement using mercury bulb thermometer considering its unsteady state behavior.	10
Q.2	Explain in detail the concept of linearization with proper example from Process Dynamics and Control	10
Q.3	Deduce the response equation for first order system when sinusoidal forcing function is given to its input.	10
Q.4	A thermometer is given the impulse change of magnitude 10. The time constant of thermometer is 6 sec. Sketch the response of thermometer and make comment on it by comparing it with step response for same system under same condition.	10
Q.5	Define offset value. Deduce an equation for offset value for unit feedback control system with P controller and servo mechanism.	10
Q.6	Two tank interacting system having time constant $\tau_1 = 0.5$, $\tau_2 = 1.5$ and $R_2 = 1$. Obtain an expression for response of liquid level in tank 2 if step change of magnitude 4 is made in the inlet flow to tank 1. Compare interacting and non-interacting system.	10
Q.7	Define transportation lag. Derive functional equations in terms of its parameter.	10
Q.8	The overall transfer function of the control system is given as $G(s) = 16 / [1.5s^2 + 2.4s + 6]$ A step change of magnitude 6 is introduced into the system. Find Decay ratio, rise time and maximum value of response.	10
Q.9	How are the selection of controllers made? Explain in detail various type of controller actions with equations for each.	10
Q.10	Explain in detail the graphical procedure followed in plotting Root Locus Diagram. Add a note on gain margin and phase margin	10



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination / Evaluation I
Academic Year: 2025-26 (semester I)

Class: TY

Course: Mechanical Operation

Course Code: PCC-CH301

Date: 22.08.2025

Time: 15.00 to 15.45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- All questions are compulsory
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Explain the term Sphericity of the particle. Prove that $\phi = \frac{6V_p}{D_p S_p}$ Consider the material is crushed uniformly in the crusher having uniform size particles then determine the total surface area of the particle. Also derive the expression for determination of total surface area of the mixture	05	CO1	BT4
	b	A crusher is reducing limestone of crushing strength 70MN/m ² from 6mm diameter size to product size of 0.1mm diameter requires 9 kw. The same machine is used to crush dolomite at the same rate of output from 6 mm diameter size to the product which consist of 20% with the average diameter of 0.25 mm, 60% with the average size diameter of 0.125 mm and the balance having average diameter of 0.085 mm. Estimate the power required to drive the crusher, assuming that the crushing strength of dolomite is 100 MN/m ² and that crushing follows Rittinngers law.	05	CO1	BT4
Q.2	a	A crushing roll 1m in diameter are set so that the crushing surfaces are 12.5 mm apart and the angle of nip is 31°. What is the maximum size of particle which should be feed to the rolls. If the actual capacity is 12% of theoretical, calculate the throughput in kg/s when running at 2.0Hz if the working face of the rolls is 0.4m long and feed weighs 2500kg/m ³	05	CO2	BT3
	b	What is screen effectiveness. Derive the expression for overall effectiveness of the screen.	05	CO2	BT3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

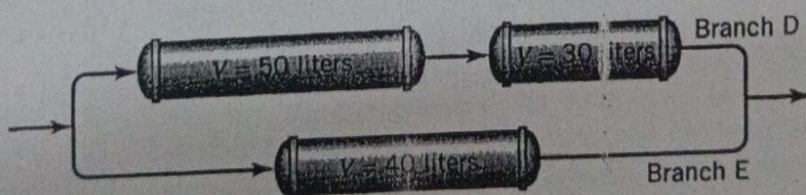
	On successful completion of this course, students should be able to:
CO 1	Understand principles of particle characterization, size reduction, and separation techniques used in process industries.
CO 2	Apply the concepts of comminution laws and screening analysis for solving particle size reduction and separation problems.
CO 3	Analyze the performance of equipment such as crushers, grinders, classifiers, cyclones, and filters in mechanical operations.
CO 4	Design and select suitable size reduction, screening, and filtration equipment for specific industrial applications.
CO 5	Evaluate the efficiency of solid-solid, solid-liquid, and solid-gas separation processes using theoretical and empirical models.
CO 6	Perform laboratory experiments on particle size analysis, filtration, sedimentation, and fluidization to validate theoretical concepts.



Date: 18/06/2025

- Note: i.) Attempt all questions
ii.) Assume suitable data if necessary iii.) Use of non-programmable calculator is permitted.
iv.) This paper contains 02 pages.

Q.No	Questions	Marks	CO	BT
Q.1	a Which is reaction rate ? Explain in brief various forms of reaction rate.	05	CH302: 1	3
	b What is irreversible reaction? Give the classification of reactions	05	CH302: 1	1& 2
	c What do you understand by elementary and non-elementary reaction? Give the right examples	05	CH302: 1	2
	d What is Arrhenius Law. Explain the relationship between temperature and activation energy.	05	CH302: 2	1& 2
Q.2	a Milk is pasteurized if it is heated to 63° C for 30 min, but if it is heated to 74°C it only needs 15 s for the same result. Find the activation energy of this sterilization process.	08	CH302: 2	3& 4
	b Give the types of reactor. What do you understand by constant volume batch reactor?	05	CH302: 3	1
	c Derive the integral equation for Irreversible Unimolecular-Type First-Order Reactions.	05	CH302: 3	3& 5
	d What do you understand by order of reaction?	02	CH302: 2	1
Q.3	a Describe the broad classification of reactor types with neat sketch	06	CH302: 3	5
	b Consider a feed $C_{A0} = 100$, $C_{B0} = 200$, $C_{I0} = 100$ to a steady-flow reactor. The isothermal gas-phase reaction is $A + 3B \rightarrow 6R$. If $C_A = 40$ at the reactor exit, what is C_B , X_A , and X_B there?	07	CH302: 4	5
	c Derive the following equation: $p_A = C_A RT = p_{A0} - \frac{a}{\Delta n} (\pi - \pi_0)$	07	CH302: 4	
Q4	a Give the material and energy balance for an elemental volume of reactor	05	CH302: 3	1
	b Derive the performance equation for plug flow reactor	05	CH302: 4	5& 6
	c The reactor setup shown in Fig.1, consists of three plug flow reactors in two parallel branches. Branch D has a reactor of volume 50 liters followed by a reactor of volume 30 liters. Branch E has a reactor of volume 40 liters. What fraction of the feed should go to branch D?	06	CH30:5	5



	d	Define space time and space velocity	04	CH305	1
					2
Q5	a	From the AH, and AH tables, I've calculated that the standard heat of my gas phase reaction at 27°C is as follows: $A + B \rightarrow 2R \quad \Delta H_{r,298K} = -50\,000 \text{ J}$ At 25°C the reaction is strongly exothermic. But this doesn't interest me because I plan to run the reaction at 1050 °C. What is the ΔHr at that temperature, and is the reaction still exothermic at that temperature? Data. Between 27°C and 1025°C the average Cp values for the various reaction components are: $\overline{C_{pA}} = 35 \text{ J/mol} \cdot \text{K}$ $\overline{C_{pB}} = 45 \text{ J/mol} \cdot \text{K}$ $\overline{C_{pR}} = 70 \text{ J/mol} \cdot \text{K}$	06	CH302: 5	5
		OR			
	a	Derive the following equation. $\Delta H_{r,2} = \Delta H_{r,1} + \int_{T_1}^{T_2} (\nabla\alpha + \nabla\beta T + \nabla\gamma T^2) dT$	06	CH302: 5	5 & 6
	b	What do you understand by recycle process? Explain in brief.	05	CH302: 5	1
	c	Describe the various contacting pattern for reactor.	05	CH302: 4	2 & 3
	d	What is order of reaction? Differentiate between Order of reaction and Molecularity	04	CH302: 1	1 & 2

Course Outcomes (CO)

- PEC-CH302:1-To get the knowledge about the basic concepts of reaction kinetics rate and order of reactions.
- PEC-CH302:2- Able to calculate material energy and behavior of the reaction balance for reactors.
- PEC-CH302:3-Able to develop and apply performance equations for batch and continuous reactors.
- PEC- CH302:4 Able to perform arrangement of reactors and accordingly evaluate the performance of reactors (sizing and volume)
- PEC-CH302:5 Develop skill to choose the right reactor among single multiple recycle etc. schemes and able to understand recycling process and the effect of selectivity and the yield of reactions

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
Summer Term Examination
Academic Year: 2024-25

Class: TY

Course: Mass Transfer II

Course Code: PCC-CH301

Date: 16.06.2025

Time: 10:00 to 13:00

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.
- Solve any 10 Questions

Q.N.			Marks	CO	BT
Q.1	a	What are the general criteria for the selection of a solvent for absorption. Explain them in detail.	5	CO1	BT2
	b	Compare the different types of membrane module.	5	CO5	BT4
Q.2	a	A gas mixture containing 3 % by volume pentane vapor and 97 % inert. The mixture is treated with a non-volatile absorption oil in an absorber and 97 % removal of pentane is required. The feed solvent is free from solute. The feed gas rate is 50 kmol/hr. The equilibrium relation is given as $Y = \frac{0.25 X}{1 + 0.75 X}$	10	CO4	BT6
Q.3	a	Define following terms 1. Absorption 2. Stripping 3. Drying 4. Free moisture 5. Absolute humidity 6. Approach in the cooling tower 7. Cooling range of cooling tower 8. Crystallization 9. Solubility 10. Membrane	10	CO4	BT2
Q.4	a	Derive the equation for height of packing for cooling tower.	10	CO1	BT6
Q.5	a	A wet solid of 28 % moisture is to be dried to 0.5 % moisture in a tray dryer. A laboratory test shows that it requires 8 hrs to reduce the moisture content of the same solid in 2 %. The critical moisture content is 6 % and the equilibrium moisture is 0.2 %. The falling rate of drying is linear in the free moisture content. Calculate the drying time of the solid if the drying conditions similar to those in the laboratory test are maintained. All moistures are expressed as % of bone-dry mass of the solid.	5	CO2	BT3

	b	Explain the moisture transport within the solid with the help of drying rate curve.	5	CO2	BT2
Q.6	a	Write down the steps to determine the number of ideal trays required for the absorption by McCabe-Thiele Method.	5	CO1	BT2
	b	Classify the dryers used in the industries.	5	CO2	BT1
Q.7	a	What is Dry-bulb and Wet-bulb temperature? Determine the relationship between them. Also, explain the wet-bulb depression.	10	CO4	BT3
Q.8	a	A solution of 30 wt % MgSO_4 is cooled to 15.6°C . What is the yield of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. Solubility is 24.5 % by weight at 15.6°C . Original mixture is 1000 kg. i. Assuming no water evaporation ii. 5 % of total water evaporation Data: Molecular weight of $\text{MgSO}_4 = 120.4$ Molecular weight of $\text{MgSO}_4 \cdot 7\text{H}_2\text{O} = 246.4$	10	CO3	BT5
	a	Explain the mechanism of crystallization.	5	CO3	BT2
Q.9	b	Explain how one can separate the Bacteria, Proteins, Lactose, Minerals and Water of milk?	5	CO4	BT6
Q.10	a	Write down the different methods of supersaturation. Also, explain Mier's Supersaturation theory in detail.	10	CO3	BT1
Q.11	a	What is a membrane? Explain the working mechanism, types, advantages, disadvantages and application of membrane separation process.	10	CO5	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Ability to estimate pressure drop, bubble size, TDH, voidage, heat and mass transfer rates for gas absorber
CO 2	Ability to scale up mass transfer processes for drying
CO 3	Calculate energy balance, yield of crystals and quantity of mother liquor in crystallizer
CO 4	Use of concept of various mass transfer operations with real life application
CO 5	Able to do separation by adsorption and design of chromatographic separation equipment's